

- 8 (a) Describe how an action potential arriving at a presynaptic neurone can result in the release of acetylcholine into the synaptic cleft.

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.....[4]

- (b) Lidocaine is a local anaesthetic drug which blocks voltage-gated Na^+ channels in the postsynaptic membrane of a synapse.

Explain the effect of lidocaine action on the postsynaptic neurone.

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.....[2]

(c) Fig. 8.1 is a diagram of a neuromuscular junction.

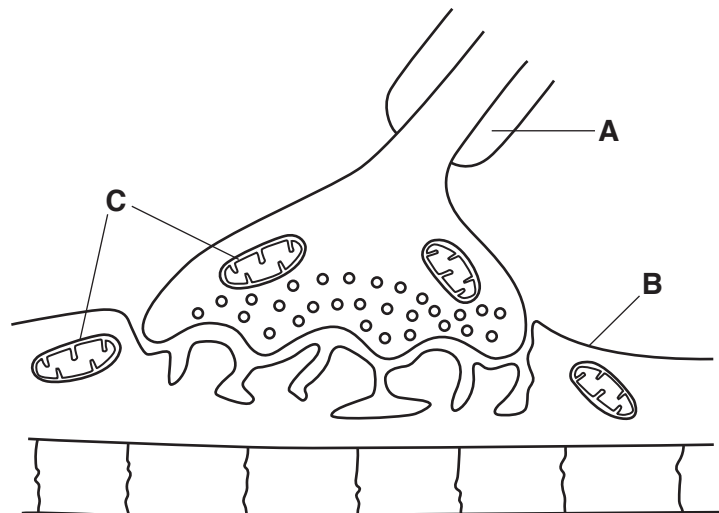


Fig. 8.1

(i) With reference to Fig. 8.1, name structures **A**, **B** and **C**.

A

B

C

[3]

(ii) Outline the importance of structure **A** in the transmission of nerve impulses.

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 [2]

(iii) Explain why the structures labelled **C** are needed in a neuromuscular junction.

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 [4]

[Total: 15]

Section B

Answer **one** question.